A CASE-STUDY BASED REPORT

ON

Movie Analytics and Box-Office Revenue Prediction System

submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

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CASE STUDIES FOR PROJECT REVIEW

Movie Analytics and Box-Office Revenue Prediction System

Project overview. This project combines string-processing algorithms, dynamic programming, graph traversal, network-flow demonstrations, recommendation logic, and an Elastic Net regression model. It uses a movie dataset covering release years 2000–2024 to explore title search, typo similarity, genre-based relationships, and retrospective worldwide revenue prediction.

CASE STUDY 1: Searching and Correcting a Movie Title

Scenario. A user wants to find a movie but enters only part of its title or misspells the name. The system should retrieve matching titles and suggest close alternatives.

Parameter	Observation
User input	Partial title or misspelled title
Algorithms	KMP and Rabin–Karp for pattern matching; Edit Distance for typo similarity
Expected output	Matching movie titles and likely correction candidates

Q1. What problem does this case address?

Answer: It helps users locate movie records even when they know only a title fragment or make a small spelling error.

Q2. What does KMP do?

Answer: Knuth–Morris–Pratt searches for a pattern in text while reusing information from previous comparisons to avoid unnecessary rechecking.

Q3. What does Rabin–Karp do?

Answer: It uses rolling hashes to quickly identify candidate pattern matches, followed by character comparison when needed.

Q4. How does Edit Distance help?

Answer: Dynamic programming computes the minimum insertions, deletions, and substitutions needed to transform one title into another; smaller distances indicate closer spellings.

Final case-study answer. The title-search module combines exact or substring search with edit-distance-based typo correction, making the dataset easier to explore through imperfect user queries.

CASE STUDY 2: Exploring Genre Connections and Revenue

Scenario. A user wants to explore movies belonging to similar genres and understand which available features are associated with worldwide box-office revenue.

Parameter	Observation
Relationship used	Shared genre tags connect movie records in a genre-based graph
Graph algorithms	Breadth-First Search (BFS) and Depth-First Search (DFS)
Prediction target	Worldwide box-office revenue
Available model inputs	Release year, parsed rating, vote count, genre tags, original language, production countries

Q1. What does the graph represent?

Answer: In this implementation, movie records are connected based on shared genre tags. It is a genre-similarity graph, not a verified actor/director collaboration graph.

Q2. What do BFS and DFS show?

Answer: BFS explores neighbouring records level by level, while DFS follows one path deeply before backtracking. Both can help traverse connected genre relationships.

Q3. Why use Elastic Net Regression?

Answer: Elastic Net combines L1 and L2 regularization with regression, helping manage multiple predictors and reducing overfitting risk.

Q4. Is the revenue prediction a pre-release forecast?

Answer: Not reliably. The available dataset does not contain budget, runtime, cast popularity, or director popularity. Rating and vote count may be known only after release, so the current model is a retrospective educational prediction.

Q5. Which columns should not be used as predictors for worldwide revenue?

Answer: Domestic revenue, foreign revenue, and revenue-share percentage fields should be excluded because they directly reveal components of the target and create target leakage.

Final case-study answer. The graph and traversal functions demonstrate relationships between genre-tagged movies, while the regression model estimates worldwide revenue from the available non-revenue fields. Results should be reported with measured test metrics rather than assumed accuracy.

MOVIE ANALYTICS — EDA FACT SHEET & VISUAL REPORT

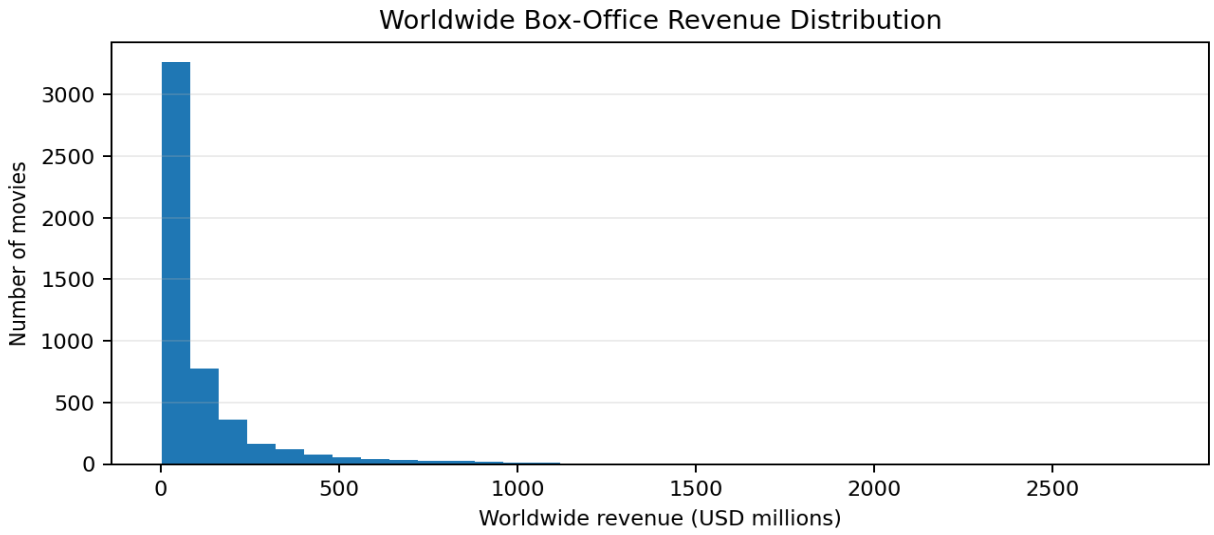
PART 1 — Dataset Overview and Data Quality

• **Dataset dimensions:** 5,000 rows × 13 columns.
• **Release years:** 2000–2024; the file contains 200 records for each year.
• **Key fields:** title, worldwide/domestic/foreign revenue, release year, genres, rating, vote count, original language, production countries.
• **Exact duplicate rows:** 0. Repeated title strings beyond their first occurrence: 45. Repeated titles are not automatically errors because titles may recur across releases.
• **Missing values:** Genres 178; Rating 170; Vote_Count 170; Original_Language 170; Production_Countries 200. Revenue and year fields are complete in this file.

Statistical Summary

Metric	Worldwide revenue	Rating	Vote count
Valid records	5,000	4,830	4,830
Mean	$119,213,693	6.51/10	2,532
Median	$48,446,575	6.60/10	1,036
Maximum	$2,799,439,100	9.70/10	36,753

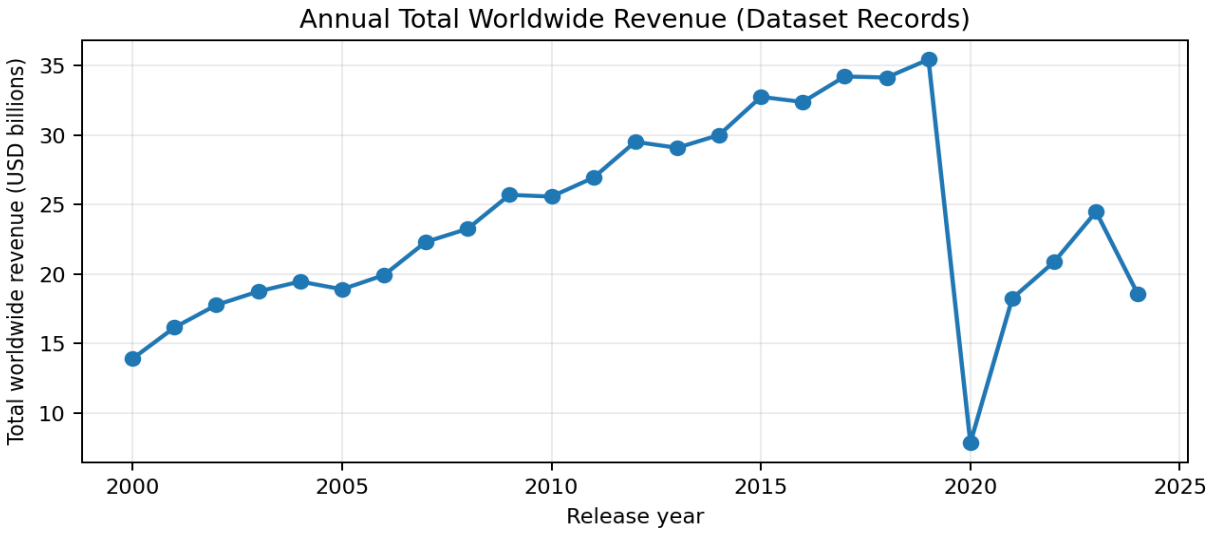
Revenue Distribution



Interpretation: Worldwide revenue is strongly right-skewed: a relatively small group of blockbusters earns very high revenue, while the median is substantially below the mean. This distribution supports considering a log-transformed target during regression.

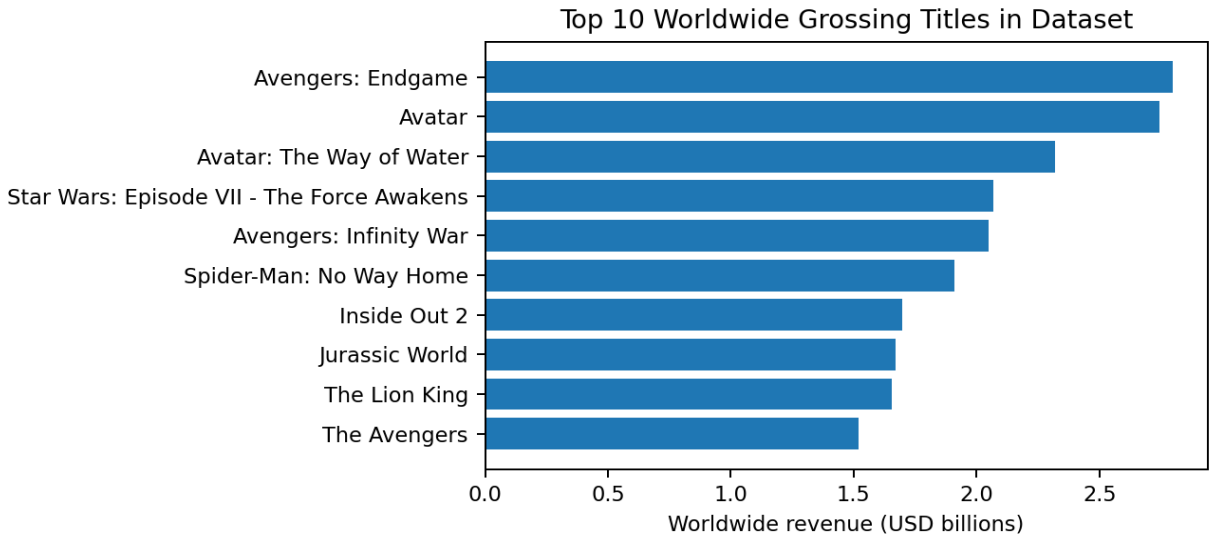
PART 1 (CONT.) — Revenue Trends and Highest-Grossing Titles

Annual Worldwide Revenue



The largest annual total in the dataset is 2019 at approximately $35.43 billion; the smallest is 2020 at approximately $7.86 billion. These totals describe this dataset's selected records, not the complete global film market.

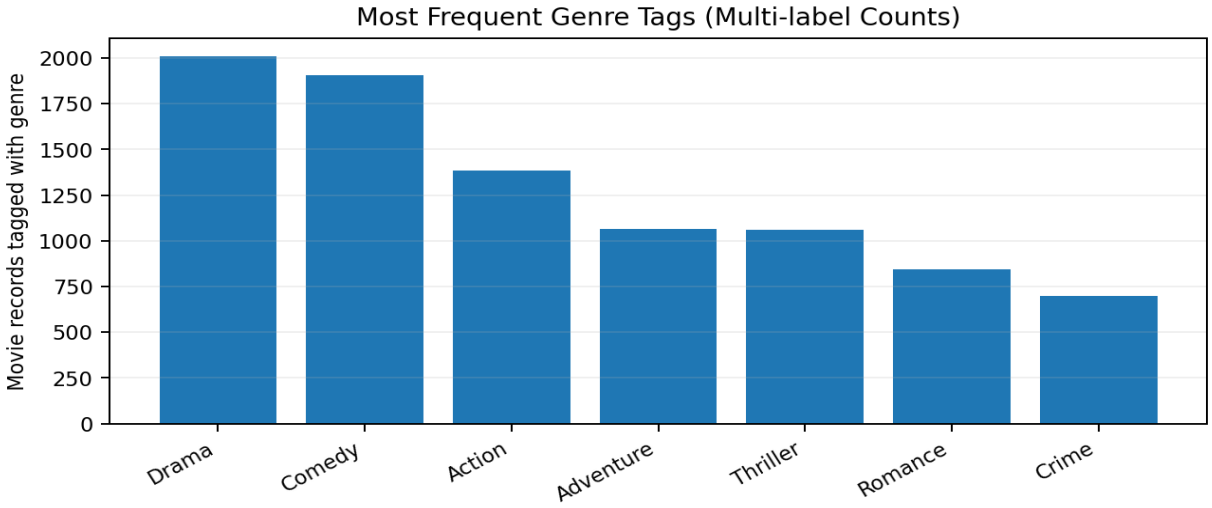
Top 10 Worldwide Grossing Titles in the Dataset



Rank	Title	Year	Worldwide revenue
1	Avengers: Endgame	2019	$2,799,439,100
2	Avatar	2009	$2,743,577,587
3	Avatar: The Way of Water	2022	$2,320,250,281
4	Star Wars: Episode VII - The Force Awakens	2015	$2,068,223,624
5	Avengers: Infinity War	2018	$2,048,359,754

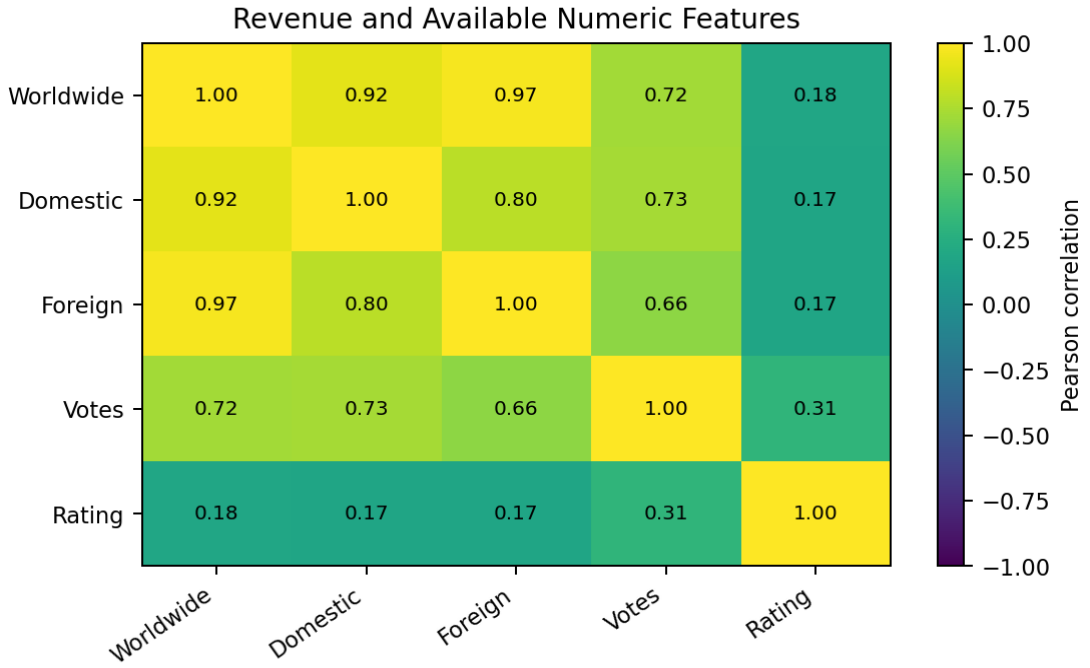
PART 1 (CONT.) — Genre, Ratings, Votes and Correlation

Most Frequent Genre Tags



Genre labels are multi-label: a movie may contribute to more than one genre count. In this dataset the most common tags include Drama, Comedy, Action, Adventure and Thriller.

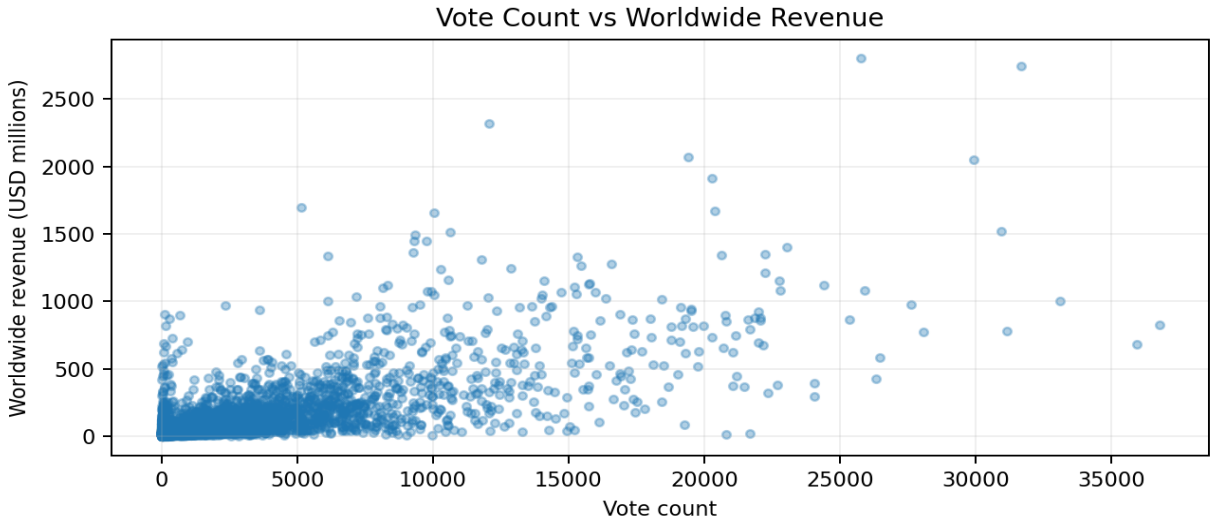
Correlation of Available Numeric Fields



Worldwide revenue has Pearson correlation of approximately 0.723 with vote count and 0.183 with rating. Domestic and foreign revenue correlate very strongly with worldwide revenue (approximately 0.915 and 0.972); these are revenue components and must be excluded from model inputs to prevent leakage.

PART 1 (CONT.) — Vote Count, Language and Country Profile

Vote Count vs Worldwide Revenue



Vote count shows a positive association with revenue in this dataset, but association does not establish causation. Vote counts can also reflect popularity after release, which limits their use for pre-release prediction.

Most Frequent Original Languages

Language	Record count
en	3,271
ja	369
zh	280
fr	203
ko	194
Missing	170
hi	92

Most Frequent Production-Country Entries

Production-country entry	Record count
United States of America	1,964
Japan	357
United Kingdom, United States of America	217
Missing	200
South Korea	180
China	172
France	124

Data-quality note: Production-country values may contain combined country strings. These counts refer to the exact strings stored in the file and should not be interpreted as a fully normalized country-level count.

PART 2 — Machine-Learning Suitability and Project Recommendations

Current Elastic Net Regression Setup

The project can use release year, rating, vote count, genre tags, original language, and production-country fields as candidate predictors. Categorical fields need encoding; numeric values need cleaning; and the worldwide revenue target may be transformed with log1p to reduce right-skew. The model must be evaluated on a held-out test set using MAE, RMSE and R^2.

Decision	Recommendation
Target	Worldwide box-office revenue; optionally train on log1p(revenue) and convert predictions back for reporting.
Leakage prevention	Do not use domestic revenue, foreign revenue, domestic %, or foreign % as predictors.
Missing values	Impute or explicitly handle missing genre, rating, vote-count, language and country fields; fit preprocessing only on training data.
Evaluation	Use a reproducible train/test split; report MAE, RMSE and R^2 from actual model execution.
Prediction scope	Call it retrospective estimation with available metadata, not pre-release forecasting.
Data representativeness	There are exactly 200 records per year; investigate how the dataset was selected before generalizing trends to the whole industry.
Feature limitations	Budget, runtime, cast popularity and director popularity are not columns in this dataset, so do not claim the current model uses ther

DSA and Algorithm Mapping

Project component	Purpose	Important limitation
KMP / Rabin–Karp	Find title patterns	Search behavior depends on normalization and query handling.
Edit Distance (DP)	Rank typo-similar titles	Thresholds are needed to avoid unrelated suggestions.
Suffix Array	Demonstrate indexed suffix-based text search	A pedagogical implementation that sorts suffixes per title/query may be expensive.
BFS / DFS	Traverse shared-genre movie graph	Graph edges represent shared genre tags, not cast/director links.
Ford–Fulkerson / Edmonds–Karp	Demonstrate maximum flow on a capacity network	Illustrative network is separate from the movie-revenue pipeline.
Elastic Net Regression	Estimate worldwide revenue	Performance must be supported by test-set metrics.

Conclusion

The dataset provides 5,000 movie records from 2000–2024 and supports a useful educational demonstration of text search, typo matching, genre-based graph traversal and regression. The strongest report will pair the visual EDA with transparent data-cleaning choices, leakage prevention, clear algorithm-to-feature mapping, and real held-out prediction metrics. Since the data are selected records and lack several production features, conclusions should be interpreted within those limitations.

Recommended workflow: load and validate data → parse revenue/rating/votes → inspect missing values and outliers → explore distributions and genres → create leakage-safe features → split data → fit preprocessing and Elastic Net → evaluate MAE/RMSE/R^2 → demonstrate search and graph algorithms.